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Sparsely sampled Fourier ptychography in diffuse reflection

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Abstract. Fourier ptychography microscopy combines the idea of aperture synthesis, which enables the synthesis of multiple low-resolution images in the frequency domain to achieve large field-of-view and high-resolution imaging results. The Fourier ptychography imaging technique applied to the field of far-field diffuse reflectance imaging needs to consider the impact of diffuse reflectance speckle formed by the object's rough surface on the imaging results. Therefore, far-field diffuse reflectance Fourier ptychography often requires more than microscopic imaging of low-resolution images to achieve super-resolution. In response, we design a sparse sampling method that can reduce the number of required low-resolution images to one-fourth of the original sampling, and does not affect the imaging resolution. Experimental results show that our method can significantly improve imaging efficiency and achieve super-resolution far-field imaging with fast sampling.

1. Introduction

In recent years, synthetic aperture technology and its ideas have been deeply rooted in the field of computational imaging research. Due to the limitation of spatial bandwidth product, it is often impossible to obtain the imaging results with both high resolution and a large field of view of the target to be measured in far-field detection. While the optical synthetic aperture technique can achieve the imaging results with high resolution and a large field of view by synthesizing a series of sub-apertures in the frequency spectrum. Fourier ptychography imaging draws on the idea of synthetic aperture^[1], using LED arrays of different positions for imaging. Acquiring multiple low-resolution images containing different spectrum components, Fourier ptychography use the phase recovery technology, and finally obtains a high-resolution, large field of view imaging results^[2-8]. The phase recovery process is essentially an inverse problem solved optimally. However, due to the limitation of the imaging distance and light source coherence of LEDs, the Fourier ptychography imaging technique cannot be applied in far-field imaging. Therefore, several universities and research institutions in the field of computational imaging have conducted imaging research on far-field Fourier ptychography.

In 2014, Dong used a highly coherent laser light source as illumination to increase the Fourier ptychography imaging distance to 1 m^[9]. Considering that the laser light source could not be used for Fourier ptychography imaging by LED array type, Dong designed an aperture scanning method for Fourier ptychography imaging scheme. Through the camera aperture movement to intercept different positions in the image spectrum, to achieve macroscopic Fourier ptychography imaging. The macroscopic Fourier ptychography here is to distinguish the Fourier ptychography microscopy imaging.



However, the imaging system designed by Dong still requires the target to be placed between the light source and the camera, and the transmissive imaging system cannot meet the needs of far-field detection.

In 2017, J.S Holloway of Rice University truly realized a reflective macroscopic Fourier ptychography imaging system^[10]. The system optical path is a modification of the transmissive optical path designed by Dong. The propagation process after the beam is irradiated to the target is mirrored and converted into a reflective optical path. Such a reflective system can achieve super-resolution imaging results of 6 times the surface contour of the object at an imaging distance of 1.2m. In addition, the work also discusses the imaging speckle on the rough target surface due to diffuse laser reflection and uses wavelet denoising to process the imaging containing speckle. Although Holloway's experimental results demonstrate that macroscopic Fourier ptychography imaging can reduce the effect of speckle on imaging. However, to achieve the same resolution improvement in microscopic imaging, the number of low-resolution images for different locations increases significantly. This undoubtedly increases the acquisition time for high-resolution imaging. In recent years, researchers have been able to achieve fast reconstruction using deep learning^[11-15]. However, the data-driven reconstruction-based approach requires a large number of samples.

To reduce the sampling requirements for target imaging in diffuse reflection Fourier ptychography imaging, and improve the reconstruction efficiency of high-resolution, large field-of-view images of the target to be measured. In this paper, we propose a sparse sampling method based on diffuse reflectance Fourier ptychography. Using the characteristics of the image spectrum in the reconstruction of diffuse reflectance objects to reduce the number of samples to one-fourth of the original number. While ensuring the overlap rate between low-resolution images. The imaging resolution of the target to be measured is not affected, and the reconstructed limit resolution is consistent with full sampling, the effectiveness of our method is demonstrated.

2. The macroscopic diffuse reflection imaging model

To achieve aperture scanning of low-resolution images, we exploit the propagation process of coherent light in far-field reflection imaging. The laser beam is emitted as highly coherent light and then modulated by the optical elements to irradiate the target to be measured. At this point, the rough surface of the target to be measured consists of a large number of irregularly distributed facets, which can usually be seen as independent diffraction units in a reflection system. Such different diffraction units add several times 2π additional phase differences to the coherent light. These different phase differences will lead to secondary interference of the reflected beam in space. In addition, due to the existence of a certain width of the point diffusion function of our imaging system. The beam reflected from a large number of facets present on the surface of the object will superimpose interference at the same image point, forming a speckle spot that distorts the image result. The specific propagation process is as follows.

For a single sub-aperture, when not considering the case of diffuse reflection noise, the laser is sent out and irradiated to the plane to be measured. On the way to be modulated by the lens for Fraunhofer diffraction, the beam will carry the object information reflected on the front surface of the lens. At this time, the light field distribution can express as follows:

$$\begin{aligned}
 U(x', y') &= \frac{e^{j\lambda d_o} e^{j\frac{k}{2d_o}(x'^2 + y'^2)}}{j\lambda d_o} \iint_{\infty} U_0(x_0, y_0) \exp\left[-j\frac{2\pi}{\lambda d_o}(x'x_0 + y'y_0)\right] dx_0 dy_0 \\
 &= \frac{e^{j\lambda d_o} e^{j\frac{k}{2d_o}(x'^2 + y'^2)}}{j\lambda d_o} F_{1/\lambda d_o}[U_0(x_0, y_0)]
 \end{aligned} \tag{1}$$

where $U(x_0, y_0)$ is object light field, $k = 2\pi/\lambda$ is the wave number, $F_{1/\lambda d_o}$ means two-dimensional Fourier transform. Coordinate scale is $1/\lambda d_o$. When there is no phase disorder caused by the rough surface, the light field distribution is the object field after a certain scale of Fourier transform and multiplied by a quadratic phase factor.

Unfortunately, the lens we use to collect target information in the far-field cannot be infinitely large. To simplify the calculation, we assume the camera lens for the ideal aberration-free thin lens. Corresponding light pupil function can be simply expressed as a circular hole, that is

$$P(u - c_u(i), v - c_v(i)) = \begin{cases} 1 & \text{if } (u^2 + v^2) < R^2 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Where R is the aperture size of the camera, as the light field on the front surface of the lens continues to propagate backward, an additional secondary phase factor $P(u, v) \exp\left[-j \frac{k}{2f}(u^2 + v^2)\right]$ is introduced through the camera lens. We use the camera to perform a two-dimensional scan in the far-field plane along the direction perpendicular to the optical axis, and the optical pupil function corresponds to each scan position $P(u - c_u(i), v - c_v(i))$. Thus the light field distribution at the camera at the position i can then be expressed as

$$\begin{aligned} U'_i(u, v) &= \frac{e^{j\lambda d_o} e^{j \frac{k}{2d_o}(u^2 + v^2)}}{j\lambda d_o} \exp\left[-j \frac{k}{2f}(u^2 + v^2)\right] P(u - c_u(i), v - c_v(i)) F_{1/\lambda d_o}[U_0(x_0, y_0)] \\ &= \frac{e^{j\lambda d_o}}{j\lambda d_o} \exp\left[j \left(\frac{k}{2d_o} - \frac{k}{2f}\right)(u^2 + v^2)\right] P(u - c_u(i), v - c_v(i)) F_{1/\lambda d_o}[U_0(x_0, y_0)] \end{aligned} \quad (3)$$

At this point and then the distance d transmitted from the rear surface of the lens, after Fresnel diffraction to reach the detector surface, to meet the imaging conditions $\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$. After two-phase modulation of the lens, we happen to eliminate the excess phase factor, and finally, the light field distribution of the detector surface can be expressed as follows

$$U_i(x, y) = \frac{e^{jk d_i}}{j\lambda d_i} \exp\left[\frac{jk}{2d_i}(x^2 + y^2)\right] F_{1/\lambda d_i} \left\{ \frac{e^{jk d_o}}{j\lambda d_o} P(u - c_u(i), v - c_v(i)) F_{1/\lambda d_o}[U_0(x_0, y_0)] \right\} \quad (4)$$

When conducting specific experiments, we need to consider the effect of diffracted image elements on the rough surface on the distribution of the optical field on the detector, but goodman et al. show in their paper that even if diffuse reflection occurs, the propagation process does not change, but only the complex amplitude distribution on the far-field Fraunhofer surface is added to the phase random function due to diffuse reflection, which is expressed as follows.

$$U(r') = \int \varphi(r_0) \exp\left[\frac{-i4\pi z(r_0)}{\lambda}\right] \exp\left(\frac{-i2\pi r_0 r'}{\lambda z}\right) dr_0 \quad (5)$$

where r_0 is the object surface coordinate, r' is the far-field surface coordinate, $\varphi(r_0)$ is the reflected light field from a smooth object surface, and $\exp(-i4\pi z(r_0)/\lambda)$ is the random phase function introduced by the speckle phenomenon, which is related to the random undulation of the rough surface. $z(r_0)$ is the height function of the rough surface.

3. The sampling of diffuse reflectance Fourier ptychography images

For macroscopic Fourier ptychography imaging in the aperture scan mode, the essence of its aperture scan is to encode and modulate the optical field to achieve super-resolution imaging. Under reflection conditions, keeping the light source and the target to be measured constant different, through the camera in the spectrum plane for ptychography scanning, the spectrum information of different positions is encoded into the camera several times to get multiple low-resolution images, the specific scanning process in the spectrum as shown in Figure.1. Among them, for the USAF resolution board shown in Figure.1(a1), the spectrum and acquisition method is shown in Figure.1(b1), Figure.1 General sampling

of USAF resolution plates and sampling of resolution plates with speckles, the USAF with/without speckle is shown as Figure.1(a1) and Figure.1(a3); the sampling of two situations is shown in Figure.1(b1) and Figure.1(b3).

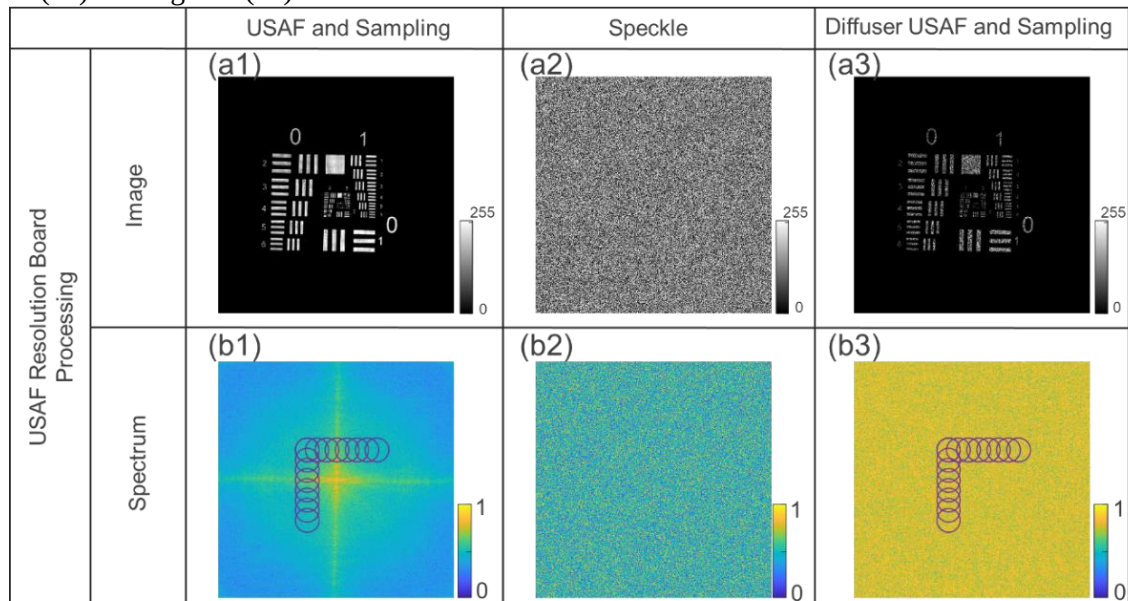


Figure 1 General sampling of USAF resolution plates and sampling of resolution plates with speckles, the USAF with/without speckle is shown as a1 and a3; the sampling of two situations is shown in b1 and b3

In the case of diffuse reflection, the spectrum of the target to be measured is modulated by the laser coherent speckle caused by diffuse reflection as a disorder distribution. The speckle and the spectrum are shown in Figure.1(a1) and Figure.1(b1). The high frequency and low-frequency information are modulated simultaneously at the position of each sub-aperture. There is no distinction between bright and dark fields during the acquisition process. This is the difference between diffuse reflectance macroscopic Fourier ptychography imaging and Fourier ptychography microscopic imaging. It is the basis of our study of a sparse sampling of diffuse reflectance Fourier ptychography imaging. Unlike the methods such as illumination angle-on-the-fly taking proposed by Tian in microscopic imaging, it can reconstruct the sub-aperture sequence by capturing only the outer side of the rectangular array while ensuring a very high overlap rate. This method discards the process of acquiring complete rectangular array sub-aperture images at high overlap rates and instead reconstructs the results by acquiring only a few layers at the periphery of the rectangular array, thus effectively reducing the number of samples and improving the imaging efficiency.

Comparing the spectrum at different locations in Figure.1, it is clear that the high-frequency information and low-frequency information of the target to be measured is randomly modulated across the spectrum due to the random speckles caused by the rough surface. That is, the frequency of the information at this point is not related to the position on the spectrum. Therefore, we take inspiration from the work of sun et al^[17]. on optimal sampling in Fourier ptychography microscopy imaging, as shown in Figure.2(b). In contrast, to complete sampling, we take only three layers of the outer rectangular array for Fourier ptychography imaging. It is worth mentioning that for objects with more complex rough surfaces, the diffuse reflections have a nearly completely random spectrum distribution, so the overlap needs to be kept above 85% to achieve sparsely sampled Fourier ptychography imaging^[18-20].

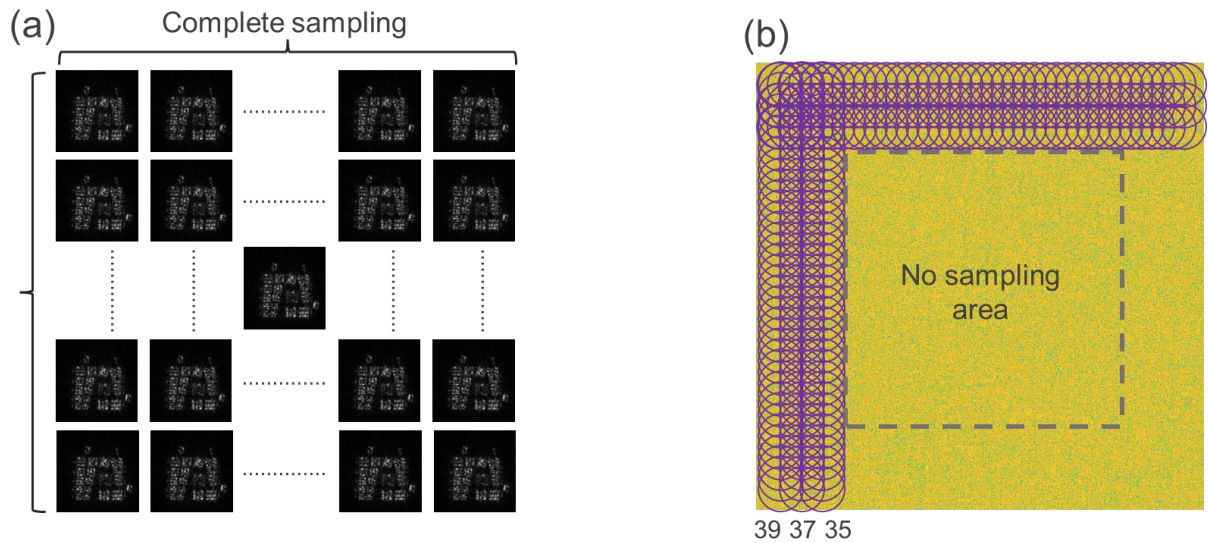


Figure 2 Diagram of the complete sampling scheme and the sampling schematic of the proposed scheme on the frequency spectrum

4. Experimental study of a sparse sampling of diffuse reflectance Fourier ptychography

In order to test the effectiveness of the proposed method, simulation reconstruction experiments are carried out using the USAF resolution plate under different sampling conditions. Figure. 3a shows the low-resolution speckle images under the limitation of diffraction. Figure.3b shows the results of reconstruction under the complete array of 1521 sub-apertures images at 39×39 . Figure.3c shows the reconstruction results obtained by the proposed method.

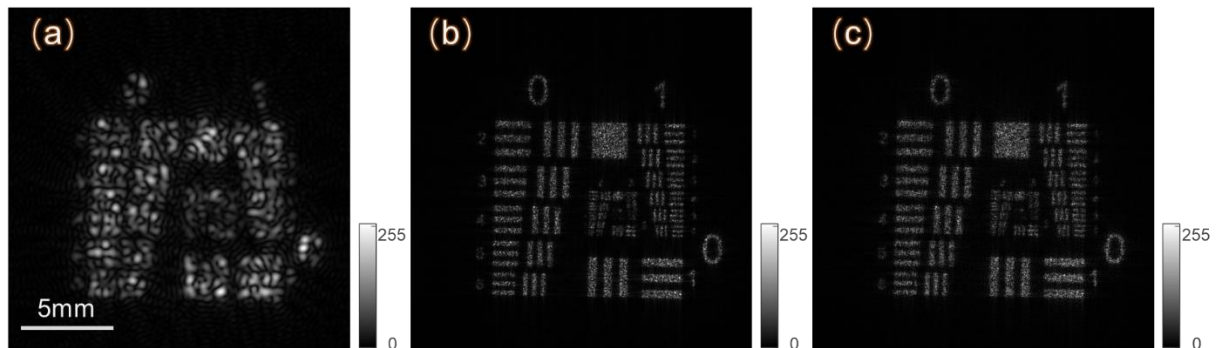


Figure 3 Result of reconstruction, the image by single aperture is shown as (a); the reconstruction by complete sampling is shown as (b); the reconstruction by our method is shown as (c).

It can be seen that the resolution of the indistinguishable region in the sub-aperture image is increased by a factor of 4 by the Fourier ptychography reconstruction, and the speckle formed by diffuse reflection is also reduced due to the increase in resolution. Figure.3(c) shows the results of the sparse sampling reconstruction of the outer three rectangular edges of the 39×39 array only, using the number of sub-apertures of 438. The high-resolution image reconstructed using sparse sampling is close to the same resolution as that of the full array reconstruction. The correctness of our proposed sampling scheme applied to the diffuse reflectance Fourier ptychography is demonstrated.

To investigate in-depth the effect of the proposed sampling scheme on the diffuse reflectance Fourier ptychography. The single aperture spectrum, Figure.4(a) and Figure.4(d), the complete sampling reconstruction spectrum, Figure.4(b) and Figure.4(f), and the spectrum of the outer three layers of sampling are compared separately (Figure.4c and Figure.4f). Due to the lack of our method relative to the complete sampling, resulting in the lack of constraints on the spectrum, the spectrum information cannot be compared as clearly as the complete sampling in general. This is the reason why we choose outer triple sampling instead of fewer samples.

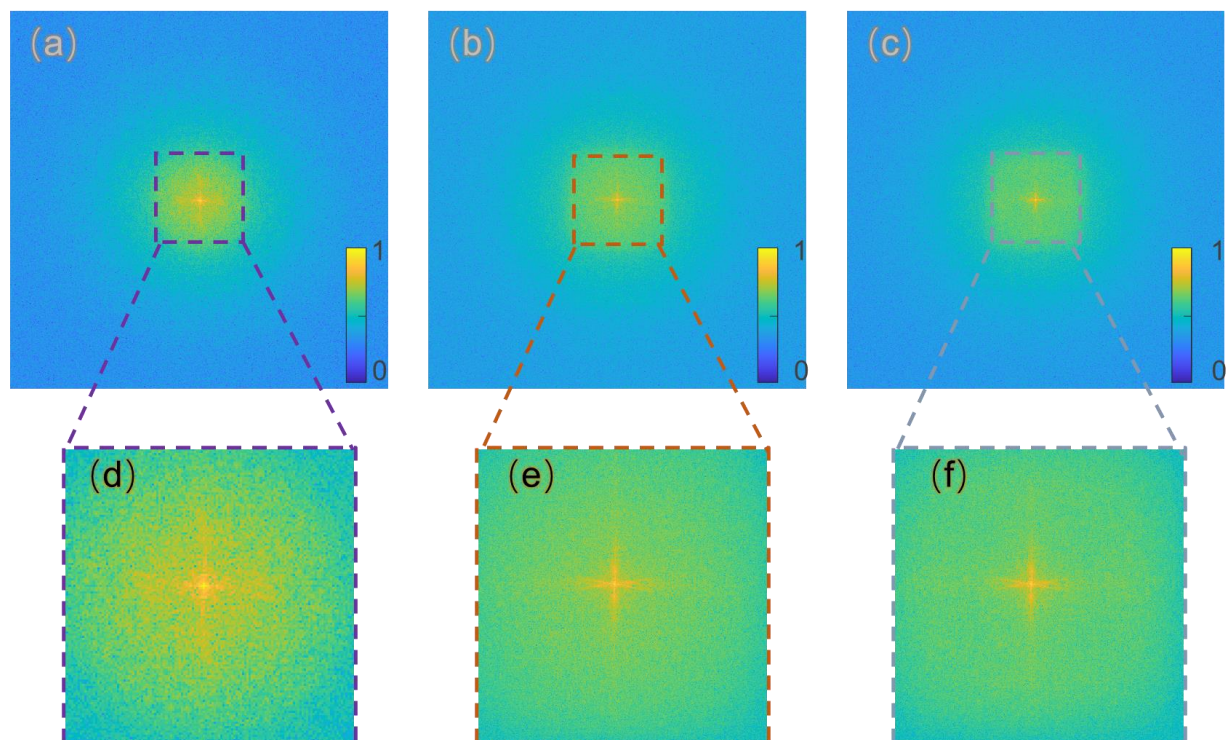


Figure 4 Single-aperture spectrum (4a), fully sampled reconstructed spectrum (4b), and spectrum of outer triple sampling we propose (4c) and corresponding zoom area

5. Conclusion

In this paper, we proposed a Fourier ptychography imaging technique based on sparse sampling for the problem of excessive sample number redundancy in reconstruction, which seriously affects the acquisition efficiency. Through the USAF resolution board simulation experiments, the same resolution improvement as the complete sampling was achieved by using sparse sampling reconstruction, and the resolvable line pairs were increased from 1 lp/mm to 4.409 lp/mm. The experiments confirmed our The experiments confirm the effectiveness of our sparse sampling scheme in diffuse Fourier ptychography. It can significantly improve the imaging efficiency and obtain high-resolution, large field-of-view imaging results.

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